

component.

Given these beliefs, the agent in question decides whether to ignore, dislike, or share the news article. If it is shared, the article moves from the agent sequentially to her connections on social media, who are then faced with the same choices. If the article is ignored or disliked, it does not get past the agent. We assume that agents receive utility when their shared content is re-shared and incur a cost when it is disliked. The former aspect captures the role of positive engagement in social media, while the latter represents the reputation loss from being called out for sharing content containing misinformation. Agents additionally receive utility from disliking (or calling out) items that they believe contain misinformation.

We characterize the Bayesian-Nash equilibria of this sequential game and prove that these equilibria always exist and are in cutoff strategies. In particular, our payoff structure implies that an individual will share any item that she believes is truthful with a high probability and will dislike articles that she believes to contain falsehoods. Items with intermediate beliefs will be ignored. Beliefs about the truthfulness of articles are formed on the basis of the article’s reliability and message, and agents’ prior beliefs/ideology. Moreover, we establish that ours is a game of strategic complements: when others are more likely to share an item, each agent also becomes more likely to do so. As a result, we show that the set of equilibria forms a lattice, with well-defined most-sharing and least-sharing equilibria. We present a number of comparative statics that describe how these extremal equilibria shift. For example, low-reliability articles are shared less, while articles that are more sensational are shared more. As expected, greater reputation costs reduce sharing uniformly.

We present two main results. First, we study the implications of the (social media) network structure. We establish non-monotone comparative statics with respect to the degree of homophily (which determines how likely agents are to be connected to others who are ideologically similar to them). Low levels of homophily ensure that agents are likely to be exposed to cross-cutting content, including “counter-attitudinal articles” that advocate views opposed to theirs. This in turn ensures that misinformation is unlikely to survive for very long. Perhaps paradoxically, for high-reliability articles, an increase in homophily reduces content virality. This is because greater homophily makes it less likely that an article escapes a given community, reducing its circulation throughout the network.

More interestingly and ominously, when relevant news items have low reliability, homophily increases virality. This is because, countering the circulation effect, high homophily also creates a perverse incentive effect: knowing that shared articles will be seen by like-minded individuals, agents become more likely to share questionable content. Strategic complementarities amplify this effect, because when others are expected to share, the benefits from sharing and reduces the likelihood of being called out for spreading misinformation are greater. It is particularly telling that homophily leads to the viral spread of low-reliability content, which are the ones more likely to contain misinformation.

We also show that political polarization and politically divisive articles are more likely to spread virally when they are low-reliability and the level of homophily is already high, generating an echo chamber-like social media environment. Strategic complementarities tend to amplify these pernicious effects of political polarization and divisive content as well.

Our second main result turns to social media platforms’ algorithm design choices and establishes